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Combining genetic and non-genetic factors to predict the risk of pancreatic cancer in patients with new-onset diabetes mellitus

Yu Zhou^{1†}, Zhuo Wu^{1,2†}, Liangtang Zeng^{1,2*} and Rufu Chen^{1,2*}

Abstract

Background Recent research suggests that new-onset diabetes mellitus (NODM) often results from pancreatic cancer (PC) rather than causing it. Determining if NODM is type 2 diabetes mellitus (T2DM) or PC-related NODM (NODM-PC) aids in the early diagnosis of PC. We developed a NODM-PC risk prediction model to stratify PC risk in patients with NODM.

Methods This study utilized data from the UK Biobank, including 238 NODM-PC cases and 14,825 cancer-free T2DM controls. Polygenic risk scores (PRSs) for PC and T2DM were constructed using previously reported single nucleotide polymorphisms (SNPs) separately, while the NODM-PC PRS was developed by combining SNPs from both. Non-genetic factors were selected as candidate predictors based on prior NODM-PC prediction models. We developed three Cox models to estimate the risk of PC diagnosis within 3 years in the NODM population and evaluated them by internal–external cross-validation.

Results Elevated NODM-PC PRS and PC PRS scores positively correlated with NODM-PC risk, while T2DM PRS showed an inverse correlation. The NODM-PC PRS achieved the highest AUC at 0.719. Three Cox models were developed: Model 1 included age at T2DM diagnosis, smoking status, HbA1c, PC PRS, and T2DM PRS; Model 2 replaced PC and T2DM PRS with NODM-PC PRS; Model 3 included only non-genetic factors. Model 2 had the highest discrimination (Harrell's C-index 0.823 (95% CI: 0.806–0.840)), demonstrated the best clinical utility with good calibration, and showed significant classification improvement (continuous net reclassification index: 26.89% and 31.93% for cases, 28.51% and 30.90% for controls, compared to Models 1 and 3). The positive predictive value for the top 1% predicted risk in Model 2 was 13.25%.

Conclusions This NODM-PC PRS enhances NODM-PC risk prediction, efficiently identifies individuals at high risk for PC screening, and improves PC screening efficiency at the population level among NODM individuals.

Keywords Pancreatic cancer, New-onset diabetes mellitus, Polygenic risk scores, Non-genetic factors

[†]Yu Zhou and Zhuo Wu contributed equally to this work.

*Correspondence:

Liangtang Zeng
zengliangtang@163.com

Rufu Chen
sychenrufu@scut.edu.cn

Full list of author information is available at the end of the article



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Background

Pancreatic cancer (PC), a leading cause of global cancer deaths, is predicted to be the second most common cause of cancer-related mortality in the USA by 2030 [1]. A primary reason for the poor prognosis of this dismal disease is that over 80% of PC cases are diagnosed at advanced stages with unresectable or metastatic lesions [2–4]. The diagnosis of PC is complicated by its asymptomatic presentation or non-specific symptoms in early stages, the lack of diagnostic biomarkers for early-stage tumors, and the pancreas's inaccessible anatomical position that hinders conventional screening strategies. This situation underscores the urgent need to develop and deploy new screening and diagnostic strategies for identifying early-stage PC patients [5].

PC has an extraordinary association with diabetes. The prevalence of diabetes mellitus (DM) or hyperglycemia (60–85%) during PC diagnosis was significantly higher than other cancers [6]. Significantly, most patients were diagnosed with diabetes within 2–3 years prior to their cancer diagnosis, a condition termed new-onset diabetes (NODM) in them [7]. The findings revealed by early data that NODM frequently abates following tumor resection suggesting that diabetes is caused by PC, and a recent causal inference research demonstrates a causal link operation from PC to NODM, but does not support a causal effect of long-standing DM on PC [8]. NODM in PC is now recognized as an early manifestation of PC and referred to pancreatogenesis or type 3c diabetes mellitus. According to a nationwide cohort study in Denmark, the 3-year cumulative incidence of PC in individuals aged 50 years and older with NODM is approximately 0.6% [9]. In other regions, this incidence can reach as high as 1%, which is significantly higher than that of the general population [10]. NODM could serve as an early potential clue for PC due to observable hyperglycemia in its asymptomatic phase. However, given the widespread prevalence of DM and the clinical indistinguishability of type 2 diabetes mellitus (T2DM) and PC-related NODM, universal screening for PC in patients with NODM is impractical. A risk-stratified screening strategy, targeting individuals at elevated risk, may optimize detection rates, decrease investigative load in those at lower risk, and potentially improve cost-effectiveness [11].

Both genetic and non-genetic factors contribute to an individual's PC risk [12]. Genome-wide association studies (GWAS) and meta-analyses have identified numerous single nucleotide polymorphisms (SNPs) associated with pancreatic cancer (PC) and type 2 diabetes (T2DM) [13–15]. Although individual variants carry a small risk with limited clinical relevance, the combined impact of multiple risk variants, as encapsulated in PRS, could enhance clinical utility for risk discrimination [16]. Combining

PRSs with non-genetic factors has been shown to significantly enhance disease prediction [17–19]. Although several risk prediction models for PC in NODM populations have been developed, most are limited to clinical data and conventional biochemical markers. No studies have yet incorporated the genetic risks of both PC and T2DM into prediction models [20–23]. This study hypothesizes that NODM resulting from PC (NODM-PC) fundamentally differs from T2DM in genetic susceptibility: NODM-PC patients have a higher genetic risk for PC, while those with new-onset T2DM likely have an increased genetic risk for T2DM itself. Consequently, we have developed a risk prediction model that uses PC and T2DM SNPs, along with clinical and demographic factors, to differentiate between T2DM and PC-induced NODM within the NODM population.

Methods

Study design and participants

The UK Biobank is a substantial biomedical database and research resource, containing detailed genetic and health information from 500,000 participants in the United Kingdom. More information about the UK Biobank can be found at: <http://www.ukbiobank.ac.uk/> (accessed in October 2023).

Our analysis focused on subjects who developed PC within 3 years post-T2DM diagnosis (NODM-PC subjects) and those diagnosed with T2DM, had over 3 years of follow-up, and remained cancer-free (Additional file 1: Fig. S1). We identified incident cases of PC and other cancers, excluding neuroendocrine tumors, within the UK Biobank cohort using national cancer registries and hospital records. T2DM identification relied on hospital records or met the diagnostic criteria with a blood glucose level (≥ 11.1 mmol/L) or glycated hemoglobin (≥ 48 mmol/mol). Diagnoses utilized the ICD- 9 and ICD-10 systems (Additional file 2: Table S1). We excluded individuals diagnosed with both T2DM and PC before enrollment. This study's data, from the UK Biobank, were downloaded in October 2023, with the last follow-up date for diagnostic outcomes being October 31, 2022. This study adheres to the guidelines of the TRIPOD statement (Additional file 3).

Ethics approval and consent to participate

The UK Biobank received ethics approval from the North West Multi-center Research Ethics Committee (REC reference: 21/NW/0157). All participants provided informed written consent, and they had the option to withdraw their personal data from the study at any time. The study utilized resources from the UK Biobank, with application number 85224.

Construction and assessment of PRS

We accessed genotype imputation data from the UK Biobank, as detailed in previous literature [24, 25]. Our approach involved using two PRSs—one for pancreatic cancer (PC PRS) and another for type 2 diabetes mellitus (T2DM PRS)—to develop a specialized PRS aimed at predicting pancreatic cancer in patients with NODM, termed the NODM-PC PRS. This specific PRS was crafted by reassessing potential associations between NODM-PC and SNPs, drawing on data from both the PC PRS and T2DM PRS, as detailed in Additional file 2: Table S2–4. The PC PRS, formulated by Sharma [15], synthesized SNPs from earlier studies, including those by Nakatochi (5 SNPs), Galeotti (30 SNPs), Molina (33 SNPs), Jia (22 SNPs), and an additional 10 SNPs linked to PC from a comprehensive pan-cancer study [8, 26–29]. For consistency with the previously reported PC PRS model, we selected 49 SNPs for our construction, excluding 5 due to strong linkage disequilibrium [15]. The T2DM PRS included 424 selected SNPs identified from genome-wide association analyses [13, 14]. The NODM-PC PRS was generated by recalculating the associations of SNPs in PC and T2DM with NODM-PC using a logistic regression model, adjusted for age at T2DM diagnosis, sex, and the first 10 principal genetic components. Ultimately, 33 potentially relevant SNPs ($p < 0.05$) (including 12 PC-related SNPs and 21 T2DM-related SNPs) were included in the NODM-PC PRS model. Association tests were performed using Plink V1.9. The published effect size, p -value, and risk allele for each SNP were sourced from the original manuscript. The weighted PRSs were generated using the PLINK “-score” command, which applies the following equation:

$$PRS = \beta_1 \times SNP_1 + \beta_2 \times SNP_2 + \dots + \beta_n \times SNP_n$$

where n represents the total SNPs count and β is the per-allele log odds ratio (OR) for PC per SNP. PRSs were standardized to a mean of zero and standard deviation (SD) of one by calculating Z scores ($\frac{PRS - \text{mean}}{SD}$). Notably, for easier validation in future studies, our clinical prediction models construction employed unstandardized PRS. To investigate the association between predicted disease risk and different PRSs, we conducted quartile analyses, considering the samples in the lowest PRS quartile as the reference group.

Non-genetic factors candidate predictors

Through a review of previously reported clinical prediction models for NODM-related PC [20, 22, 30], we identified the following variables as candidate predictors: age at diagnosis of T2DM, sex, BMI, smoking, glucose, glycated hemoglobin (HbA1c), hemoglobin concentration,

triglycerides, alanine aminotransferase (ALT), creatinine, platelet count, cholesterol, and alkaline phosphatase. The specific fields for variable selection are listed in Additional file 2: Table S5. However, due to the lack of detailed records in the UK Biobank for variables such as weight changes, glucose changes, BMI changes, as well as symptoms within 6 months before the prediction date (e.g., abdominal pain, weight loss, jaundice, heartburn, indigestion, and nausea), these variables were not included as candidate predictors in this study.

Missing data

Variables such as BMI, smoking, glucose, HbA1c, hemoglobin concentration, triglycerides, ALT, creatinine, platelet count, cholesterol, and alkaline phosphatase had missing values. Under the assumption of missing at random (MAR), multiple imputation by chained equations (MICE) was applied to generate 10 imputed datasets. Additional file 2: Table S6 summarizes the extent of missing data for the candidate predictor variables. The imputation model included the outcome, all candidate predictors, and the Nelson–Aalen cumulative hazard estimate [31]. BMI was log-transformed to ensure normality during imputation and subsequently back-transformed for modeling. The imputed datasets were used for model fitting and subsequent analyses.

Model development and evaluation

This study adopted an analytical strategy of fitting the model to the entire study cohort and evaluating its performance using internal–external cross-validation (IECV) [32]. In the IECV process, the dataset was non-randomly divided based on the geographic regions of the UK Biobank. In each iteration, the model was trained on data from all regions except one, and its performance was evaluated on the excluded region. Regions with smaller sample sizes were merged into neighboring regions, resulting in a total of 10 regions (see Additional file 2: Table S7). This process was repeated for each region to generate predictions for all excluded individuals. Unlike a single random split, IECV provides a more robust evaluation of the model's performance and applicability to new samples. By utilizing the entire dataset for training and testing, IECV ensures a comprehensive assessment of the generalizability of prediction models in large datasets [33, 34].

Restricted cubic spline (RCS) curves were used to explore potential nonlinear associations between continuous variables and PC. Using the complete case data, the best functional forms for continuous variables were explored using fractional polynomials (FPs) with up to two powers [35]. FPs are a flexible approach to modeling non-linearities in continuous variables, and “two powers”

refers to up to two coefficients being used to model the variable–outcome relationship, potentially raised to powers that are integers or fractions. These FP terms were used in the Cox modeling.

To compare the improvements achieved by incorporating different genetic risk scores into the non-genetic factor model, we developed three Cox proportional hazards models. Model 1 combined non-genetic factors with PC PRS and T2DM PRS. Model 2 was constructed by replacing the PC PRS and T2DM PRS in Model 1 with the NODM-PC PRS. Model 3 was constructed using only non-genetic factors. The prediction model was fitted using candidate predictors selected for each model, including pre-specified interactions between age and sex.

Continuous variables and interactions associated with $p < 0.05$, and categorical variables associated with an exponentiated coefficient (HR) > 1.1 or < 0.9 with $p < 0.05$ were selected for inclusion in the final model where they were clinically plausible. This predictor selection approach considers both the statistical and clinical significance of predictor–outcome associations. Rubin's rules were used to combine coefficients across the imputed datasets.

The predicted risks from the Cox model can be obtained by combining the linear predictor with the baseline hazard function, using the formula: predicted event probability $= 1 - \text{St}^{\exp(X\beta)}$, where St represents the baseline survival function at 3 years, and $X\beta$ is the linear predictor. Region-level estimates of Harrell's C-index, calibration slope, and calibration-in-the-large for each model were combined using random effects meta-analysis with the Hartung-Knapp-Sidik-Jonkman method, based on the individual-level predictions generated during IECV. The pooled predictions generated from IECV were utilized to generate smoothed calibration plots and conduct decision curve analysis and precision recall (PR) curves, enabling a comparison of the clinical utility across all models. For each model, sensitivity, specificity, and positive predictive value (PPV) were assessed based on using cut-offs at the highest 1%, 5%, and 10% of their predicted risk distributions. In addition, to compare the improvement in the model by including different PRS, we calculated the continuous net reclassification improvement (NRI), categorical NRI, and the associated integrated discrimination improvement (IDI) for each model at prediction probability thresholds of 0.50%, 1%, 5%, and 10%.

Sample size considerations

Before the study, we determined that to minimize model overfitting (a target shrinkage factor of 0.9) and ensure precise estimation of overall risk, we required a minimum sample size of 7954 participants (125 outcomes)

based on a maximum of 20 parameters, Cox-Snell R^2 of 0.02235 (15% of the maximum permitted value, 0.149), estimated outcome rate of 0.0053 per person year, and a 3 year time horizon, and a mean follow-up period of 2.96 years [36].

Statistical software

All statistical analyses were conducted using R software version 4.2.2 and SPSS v 26.0. All p -value < 0.05 (two-sided) was considered significant.

Results

Baseline characteristics

The study cohort comprised 15,063 NODM individuals, with 238 diagnoses of PC in 44,602.7 person-years, resulting in a crude incidence rate of 5.3 per 1000 person-years. The median age at diagnosis of T2DM in the NODM-PC case group was significantly higher at 70 years (interquartile range (IQR): 66–74), compared to 64 years (IQR: 57–70) in the control group. In terms of sex distribution, both groups were very similar, with approximately 40% female and 60% male in both the control and case groups. Details on baseline characteristics of both datasets are available in Additional file 2: Table S6.

Construction of PRSs model

As described in the Methods, the PC PRS was developed to evaluate PC risk in the NODM population, whereas the T2DM PRS was designed to assess T2DM probability in the same group. We then created a NODM-PC PRS by merging SNPs from both diseases to enhance PC risk prediction in the NODM group. The density plots of the scores for each PRS were shown in Fig. 1. For both PC PRS and NODM-PC PRS, scores shift towards higher values in NODM-PC cases than in cancer-free T2DM controls, indicating increased risk. This trend reverses for the T2DM PRS (Fig. 1A).

The associations between PRSs and NODM-PC risk across four score quartiles are shown in Table 1. A high PC PRS is associated with an increased risk of NODM-PC, whereas a high T2DM PRS is associated with decreased risk. Notably, the HRs for NODM-PC were significantly elevated in the top two quartiles of PC PRS, in contrast to significantly reduced HRs in the top two quartiles of T2DM PRS. The combined NODM-PC PRS, incorporating SNPs from both PC PRS and T2DM PRS, demonstrates a more robust correlation with the risk of NODM-PC. Patients in each quartile of NODM-PC PRS exhibit notably higher HRs compared to those in the first quartile. In keeping with this, the AUC observed in the NODM-PC PRS of 71.9% (95% CI: 68.7–75.1%) was significantly higher than that of the PC PRS ($p < 0.001$) and the T2DM PRS ($p < 0.001$).

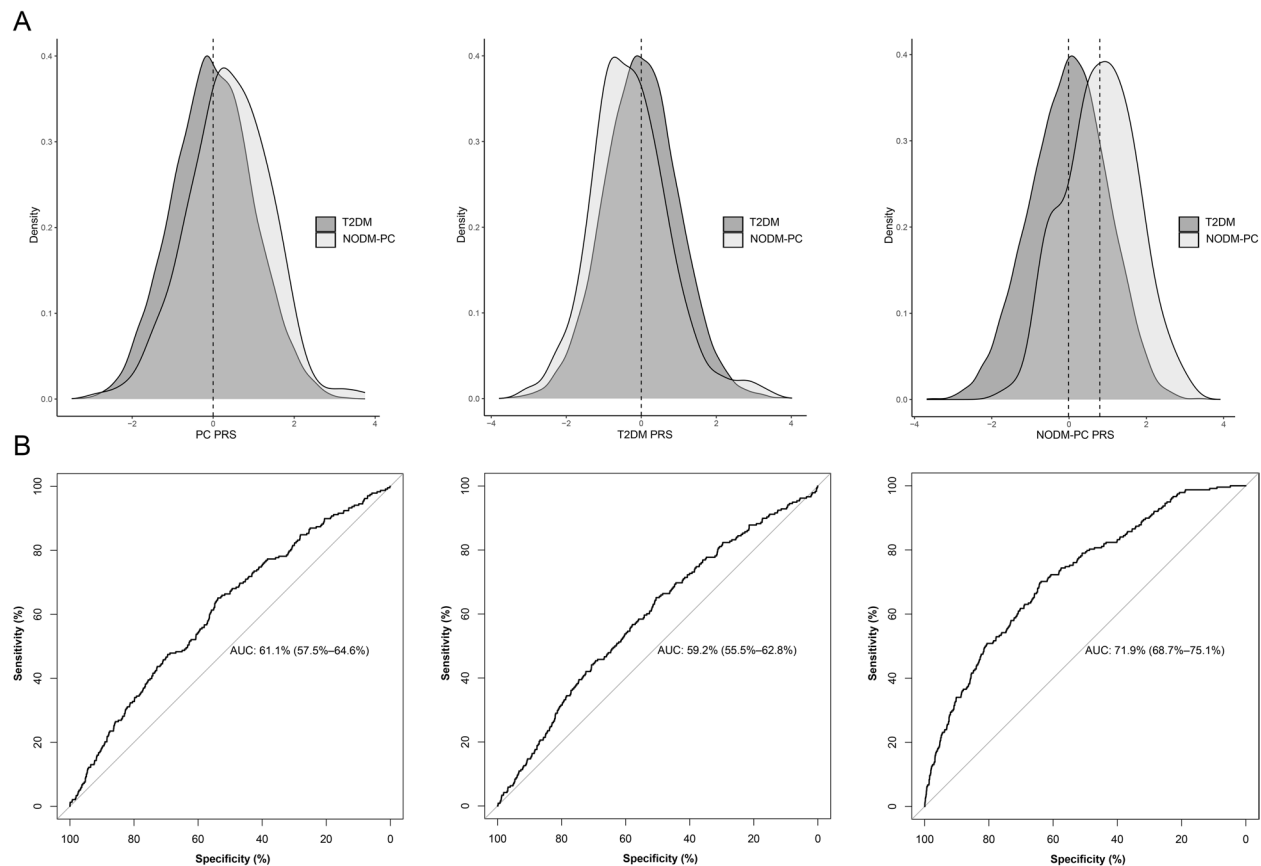


Fig. 1 Distribution and performance of PRS. **A** Density plots illustrate the distribution of standardized PRS among NODM-PC cases and cancer-free T2DM controls; **B** ROC curves, along with AUC values and 95% confidence intervals, demonstrate the overall accuracy of NODM-PC prediction for each PRS

Table 1 Association between PRSs and risk of NODM-PC in UK Biobank cohort

Quintile	No. cases	No. controls	HR (95% CI) ^a	P	HR (95% CI) ^b	P
PC PRS						
1	32	3737	Reference	NA	Reference	NA
2	48	3717	1.50 (0.96–2.35)	0.073899	1.40 (0.89–2.19)	0.141077
3	64	3699	2.01 (1.32–3.07)	0.001268	1.90 (1.24–2.92)	0.00308
4	94	3672	2.96 (1.98–4.42)	1.15E – 07	2.95 (1.96–4.43)	1.93E – 07
T2DM PRS						
1	90	3676	Reference	NA	Reference	NA
2	65	3701	0.72 (0.52–0.99)	0.043406	0.76 (0.55–1.04)	0.084279
3	45	3720	0.50 (0.35–0.71)	0.000129	0.56 (0.39–0.81)	0.001665
4	38	3728	0.42 (0.29–0.61)	0.000007	0.50 (0.34–0.75)	0.000668
NODM-PC PRS						
1	15	3751	Reference	NA	Reference	NA
2	35	3732	2.34 (1.28–4.28)	0.005933	2.26 (1.23–4.14)	0.00857
3	59	3706	3.96 (2.25–6.97)	0.000002	3.84 (2.17–6.79)	0.000004
4	129	3636	8.71 (5.11–14.87)	2.09E – 15	8.49 (4.94–14.58)	9.16E – 15

^a Model was not adjusted

^b Model was adjusted for age at diagnosis of T2DM, sex, and the first 10 principal components of ancestry

Model development

The RCS analysis revealed a significant nonlinear association for HbA1c (Additional file 1: Fig. S2). In the development of the Cox models, we selected nonlinear (FP) terms for HbA1c. Based on the above results, we developed three models. Model 1 includes PC PRS, T2DM PRS, age at diagnosis of T2DM, smoking status, HbA1c. Model 2 is equivalent to replacing PC PRS and T2DM PRS with NODM-PC PRS. Model 3 was constructed using only non-genetic factors. Detailed information about the models can be found in Additional file 2: Tables S8–10.

Model performance

Figure 2 presents the region-level and meta-analysis pooled estimates of Harrell's C-index for all models. The incorporation of PRS into non-genetic factors models significantly enhances the models' discriminatory capability. Model 2 achieved the highest predictive performance, with Harrell's C-index of 0.823 (95% CI, 0.806–0.840), compared to Model 1 (0.782, 95% CI: 0.747–0.818) and Model 3 (0.749, 95% CI: 0.712–0.786). Among the three models, Model 2 was well calibrated on summary metrics, with a calibration slope of 0.994 (95% CI: 0.885–1.102) and a calibration in the large of -0.006 (95% CI: -0.115 – 0.102) (Additional file 1: Fig. S3–4). The smoothed calibration curves for all models and their summary measures are shown in Fig. 3. The precision-recall (PR) curves for the three models are presented in Additional file 1: Fig. S5. Model 1, Model 2, and Model 3 achieved a PR-AUC of 5.84%, 7.98%, and 4.61%, respectively. Among the three models, Model 2 consistently demonstrated the best performance in terms of precision-recall analysis.

Models PPV and clinical utility

In Model 2, the PPVs for the top 1%, 5%, and 10% of risk predictions using IECV were 13.25%, 9.55%, and 7.37%, respectively. These values were higher compared to Model 1 and Model 3 (Additional file 2: Table S11). Specifically, to screen the top 1% of the high-risk population, achieving a sensitivity of 8.40% and a specificity of 99.12%. In addition, decision curve analysis demonstrated that Model 2 consistently provided a greater net benefit compared to both Model 1 and Model 3 for threshold probabilities below 10% (Fig. 4). Furthermore, Model 2 outperformed the “treat all” strategy, supporting its potential utility in guiding decisions for targeted screening studies in high-risk populations.

NRI and IDI analysis

The inclusion of NODM-PC PRS to the non-genetic model improved NODM-PC classification accuracy

for NODM-PC. The continuous net reclassification improvement (NRI) for cases showed an overall increase of 31.93% (95% CI, 20.65%, 43.64%), while for controls, it increased by 30.90% (95% CI, 29.26%, 32.53%) (Additional file 2: Table S13). In comparison, the inclusion of PC PRS and T2DM PRS increased the continuous NRI to 16.81% (95% CI, 3.94%, 28.12%) for cases and 16.61% (95% CI, 14.96%, 18.15%) for controls (Additional file 2: Table S12). Categorical NRIs are presented in Additional file 2: Table S12–13.

According to the integrated discrimination improvement (IDI) metric, after the incorporation of NODM-PC PRS into the non-genetic model, the increase in risk difference between cases and non-cases was 2.11% (95% CI, 1.46–2.76%) (Additional file 2: Table S13). In contrast, the integration of PC PRS and T2DM PRS increased the IDI to 0.59% (95% CI, 0.28–0.90%) (Additional file 2: Table S12). Additional file 2: Table S14 provides comparison results of the NRI and IDI between Model 2 (which incorporates NODM-PC PRS and the non-genetic model) and Model 1 (which incorporates PC PRS, T2DM PRS and the non-genetic model).

Discussion

Our study of NODM-PC cases and T2DM controls from the UK Biobank found that the PC PRS is associated with a higher NODM-PC risk, similar to the general population. We also discovered an inverse relationship between T2DM PRS and NODM-PC risk in NODM patients, indicating distinct genetic factors for NODM from pancreatic cancer versus typical T2DM. Moreover, we created a new PRS model for NODM-PC that outperforms PC and T2DM PRS in differentiating NODM-PC from new-onset T2DM. Based on these results, we developed and compared three Cox models. The results show that Model 2, which integrates the NODM-PC PRS with non-genetic factors, performed best in terms of discrimination, calibration, and net benefit. Compared to Models 1 and 3, it significantly improved the identification and reclassification of NODM-PC cases.

PC remains one of the dismal malignancies. Over the years, multimodal treatments for PC, including chemotherapy, surgical resection, and radiation therapy, have been extensively studied [37]. Currently, the treatment modalities for resectable PC have greatly improved patient prognoses, making early diagnosis essential for enhancing patient outcomes [38]. Previous studies have reported early risk prediction models for PC in NODM populations. Hsieh et al. [39] developed a PC prediction model for T2DM, with an AUC of only 0.727; however, the study did not specifically focus on NODM-PC. The “END-PAC” 3-variable score, developed with 64 PC cases and validated with only 9 events [30], was

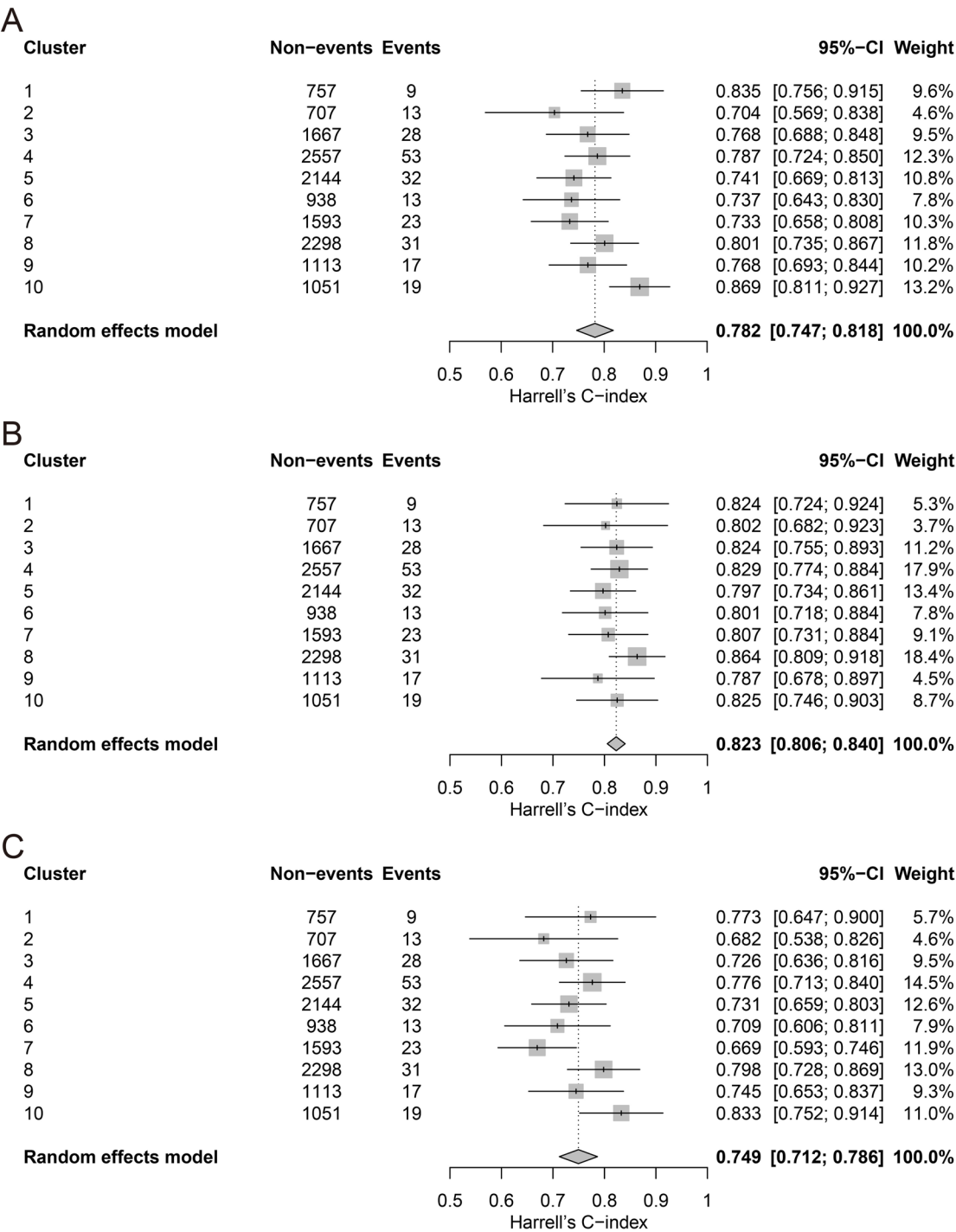


Fig. 2 Regional-level and overall meta-estimates of Harrell's C-index for Model 1 (A), Model 2 (B), and Model 3 (C). Note: The geographic region corresponding to "Cluster" in the figure is provided in Additional file 2: Table S7

externally validated using larger datasets (up to 99 cases) [40], achieving an AUC of 0.75 and a positive predictive value of 2.0%. Cichosz et al. [21] constructed a machine learning model incorporating age, gender,

and routine biochemical variables to effectively differentiate PC-related diabetes, with an AUC of 0.78. Clift et al. [20] proposed a Cox regression-based model relying on clinical parameters to assess PC risk in adults

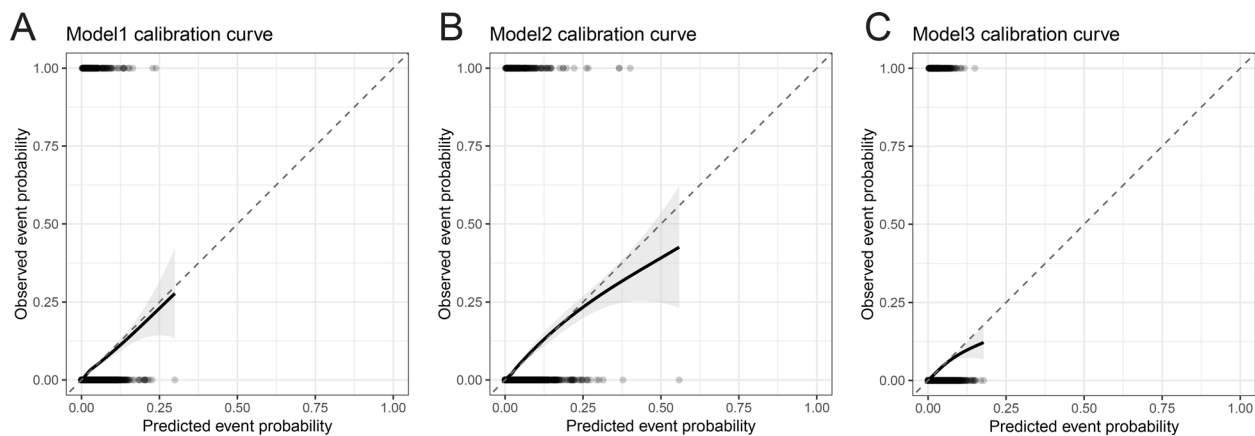


Fig. 3 Calibration curves for Model 1 (A), Model 2 (B), and Model 3 (C)

with NODM, achieving a C-index of 0.802. However, none of these studies incorporated genetic background into their models.

Although this study is not the first to develop a PC risk prediction model for the NODM populations, we have innovatively incorporated genetic risk as a key predictor, taking into account the genetic background of both PC and T2DM to more accurately distinguish NODM-PC from T2DM.

Since the publication of the first cancer genome-wide association study, thousands of common alleles associated with cancer risk have been identified [16]. Research has identified numerous SNPs associated with the risks of PC, and corresponding PRS have been developed, significantly enhancing the performance of predictive models [8, 15, 26–29]. Taking into consideration the fundamental pathological and physiological differences between PC-related NODM and true T2DM, our study for the

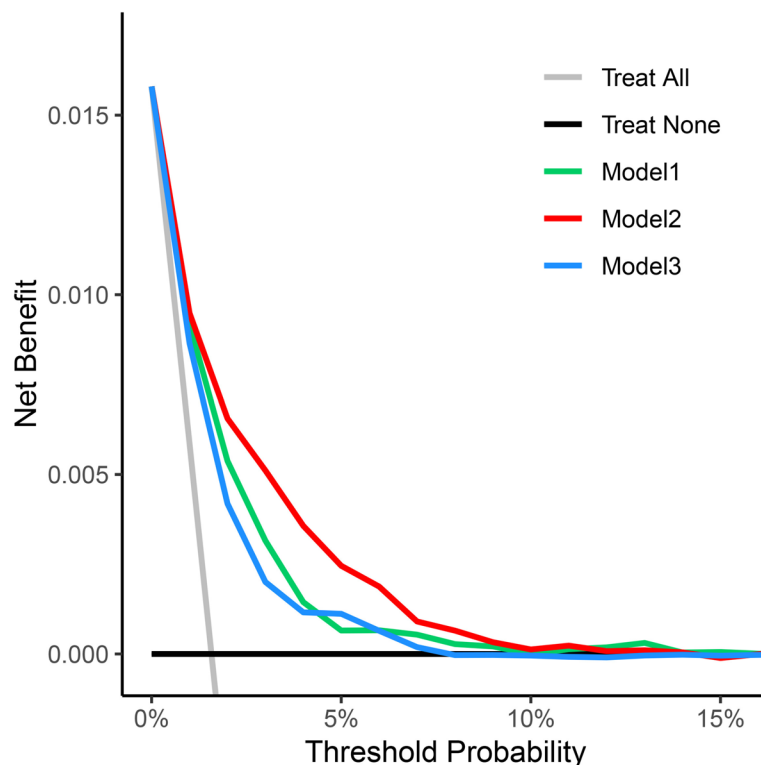


Fig. 4 Decision curve analysis comparing the clinical utility (net benefit) of each model developed, compared to “treat all” and “treat none” strategies

first time contemplated incorporating the genetic risk of T2DM into the predictive scoring model for NODM-PC. Our NODM-PC PRS model, combining risk SNPs from PC and T2DM, shows enhanced predictive power. The final model, merging NODM-PC PRS scores with clinical demographics, achieves a Harrell's C-index of 0.823, aiding early PC detection in groups lacking screening guidelines. By identifying high-risk NODM individuals, our model improves the cost-effectiveness of further clinical screenings.

The strength of our study lies in its use of the UK Biobank, a prospective cohort, which minimizes confounding bias. Unlike previous research predicting PC risk, our study is the first to include genetic risk factors beyond PC-specific SNPs. We combined genetic risks associated with both T2DM and PC, which outperforms models based solely on PC SNPs and highlights the important value of incorporating broader genetic risk factors in risk stratification.

This study has several limitations. First, there is the potential non-representativeness of the UK Biobank for the entire British population, given that its participants are generally health-conscious and well-educated. In future studies, methods such as weighting might be considered to address this challenge [41]. The cohort's age bias towards those over 40 may underrepresent younger individuals, although PC is rare in people under 40. Second, the UK Biobank primarily comprises individuals of European ancestry, underscoring the need for further research to assess the impact of these SNPs on other ethnic groups. Third, due to the lack of data in the UK Biobank on weight changes, glucose changes, and BMI changes at the time of T2DM diagnosis, these variables were not included as candidate predictors in this study. Fourth, as the data in this study were entirely derived from the UK Biobank, differences in geography, sample sources, and dataset characteristics may result in a PC incidence among NODM patients that differs from other populations. This could potentially affect the PPV of the model, particularly in settings with lower or higher baseline PC prevalence. Despite our NODM-PC PRS model's effectiveness in identifying high-risk individuals within the NODM population, external validation is still needed.

Conclusions

In summary, we developed and validated a prediction model for NODM-PC to identify individuals at high risk for PC within the NODM population. This highlights the significance of PRS in enhancing early diagnostic models for NODM-PC. The inclusion and identification of additional T2DM-related risk SNPs in the model could prove beneficial, and further development of NODM-PC PRS may lead to significant enhancements.

Conducting GWAS in larger cohorts for identifying relevant SNPs and developing a robust NODM-PC PRS is essential. As GWAS costs decrease, expanding validation samples and using external datasets for validation will help confirm our models' effectiveness and reliability. These findings offer crucial insights for future NODM-PC clinical prediction model development.

Abbreviations

NODM	New-onset diabetes mellitus
PC	Pancreatic cancer
T2DM	Type 2 diabetes mellitus
PRS	Polygenic risk score
SNPs	Single nucleotide polymorphisms
GWAS	Genome-wide association study
ICD- 9	International Classification of Diseases- 9
ICD- 10	International Classification of Diseases- 10
SD	Standard deviation
ROC	Receiver operating characteristic curve
AUC	Area under the curve
NRI	Net reclassification improvement
IDI	Integrated discrimination improvement
HR	Hazard ratio
CI	Confidence interval
PPV	Positive predictive value

Supplementary Information

The online version contains supplementary material available at <https://doi.org/10.1186/s12916-025-04048-4>.

Additional file 1. Figure S1. Selection of NODM-PC cases and cancer-free T2DM controls from the UK Biobank. Figure S2. Analysis of Restricted Cubic Spline (RCS) Curves for Continuous Variables. Figure S3. Regional-level and overall meta-estimates of calibration slope estimates for Model 1 (A), Model 2 (B), and Model 3 (C). Figure S4. Regional-level and overall meta-estimates of calibration-in-the-large estimates for Model 1 (A), Model 2 (B), and Model 3 (C). Figure S5. Precision-Recall curve of different models.

Additional file 2. Table S1. Data fields and International Classification of Disease Codes used for identification of PC and T2DM in the UK Biobank cohort. Table S2. Information of 424 SNPs for T2DM. Table S3. Information of 49 SNPs for PC. Table S4. Information of 471 SNPs for NODM-PC in the UK Biobank. Table S5. Information on the data field for variables in the UK Biobank. Table S6. Baseline characteristics of the study population, NODM-PC cases, and cancer-free T2DM controls. Table S7. Geographic region divisions of assessment centers in the UK Biobank. Table S8. Coefficients for variables in model1. Table S9. Coefficients for variables in model2. Table S10. Coefficients for variables in model3. Table S11. Models diagnostic performance at different predicted risk cut-offs. Table S12. Risk reclassification analysis at 0.5%, 1%, 5%, and 10% predicted probability cut-offs for NODM-PC: Comparison of recalibrated model1 and model3. Table S13. Risk reclassification analysis at 0.5%, 1%, 5%, and 10% predicted probability cut-offs for NODM-PC: Comparison of recalibrated model2 and model3. Table S14. Risk reclassification analysis at 0.5%, 1%, 5%, and 10% predicted probability cut-offs for NODM-PC: Comparison of recalibrated model2 and model1.

Additional file 3. TRIPOD Checklist: Prediction Model Development and Validation.

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Authors' contributions

R.C. and Y.Z. conceptualized and designed the study. L.Z. and Z.W. performed the data analysis. L.Z., Y.Z. and Z.W. drafted the manuscript. L.Z. and Z.W. contributed to the analysis and interpretation of data. R.C., Y.Z., Z.W. and L.Z. contributed to revision of the manuscript. R.C. and L.Z. were involved in study supervision. All authors read and approved the final version of the manuscript.

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Data availability

The data that support the findings of this study are available from UK Biobank but restrictions apply to the availability of these data, which were used under license for the current study, and so are not publicly available. This research was carried out using the UK Biobank Resource under application number 85224.

Declarations

Ethics approval and consent to participate

The UK Biobank received ethics approval from the North West Multi-center Research Ethics Committee (REC reference: 21/NW/0157). All participants provided informed written consent, and they had the option to withdraw their personal data from the study at any time. The study utilized resources from the UK Biobank, with application number 85224.

Consent for publication

Not applicable.

Competing interests

The authors declare no competing interests.

Author details

¹Department of Pancreatic Surgery, Department of General Surgery, Guangdong Provincial People's Hospital (Guangdong Academy of Medical Sciences), Southern Medical University, Guangzhou, Guangdong Province, China.

²School of Medicine, South China University of Technology, Guangzhou, Guangdong Province, China.

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References

- Rahib L, Wehner MR, Matrisian LM, Nead KT. Estimated projection of US cancer incidence and death to 2040. *JAMA Netw Open*. 2021;4(4):e214708.
- Ben-Ami R, Wang QL, Zhang J, Supplee JG, Fahrman JF, Lehmann-Werman R, et al. Protein biomarkers and alternatively methylated cell-free DNA detect early stage pancreatic cancer. *Gut*. 2024;73(4):639–48.
- Ryan DP, Hong TS, Bardeesy N. Pancreatic adenocarcinoma. *N Engl J Med*. 2014;371(11):1039–49.
- Kleeff J, Korc M, Apte M, La Vecchia C, Johnson CD, Biankin AV, et al. Pancreatic cancer. *Nat Rev Dis Primers*. 2016;2:16022.
- Halbrook CJ, Lyssiotis CA, Pasca di Magliano M, Maitra A. Pancreatic cancer: advances and challenges. *Cell*. 2023;186(8):1729–1754.
- Aggarwal G, Kamada P, Chari ST. Prevalence of diabetes mellitus in pancreatic cancer compared to common cancers. *Pancreas*. 2013;42(2):198–201.
- Sah RP, Nagpal SJ, Mukhopadhyay D, Chari ST. New insights into pancreatic cancer-induced paraneoplastic diabetes. *Nat Rev Gastroenterol Hepatol*. 2013;10(7):423–33.
- Molina-Montes E, Coscia C, Gómez-Rubio P, Fernández A, Boenink R, Rava M, et al. Deciphering the complex interplay between pancreatic cancer, diabetes mellitus subtypes and obesity/BMI through causal inference and mediation analyses. *Gut*. 2021;70(2):319–29.
- Jensen MH, Cichosz SL, Hejlesen O, Henriksen SD, Drewes AM, Olesen SS. Risk of pancreatic cancer in people with new-onset diabetes: a Danish nationwide population-based cohort study. *Pancreatol*. 2023;23(6):642–9.
- Chari ST, Leibson CL, Rabe KG, Ransom J, de Andrade M, Petersen GM. Probability of pancreatic cancer following diabetes: a population-based study. *Gastroenterology*. 2005;129(2):504–11.
- Pashayan N, Morris S, Gilbert FJ, Pharoah PDP. Cost-effectiveness and benefit-to-harm ratio of risk-stratified screening for breast cancer: a life-table model. *JAMA Oncol*. 2018;4(11):1504–10.
- Zeng L, Wu Z, Yang J, Zhou Y, Chen R. Association of genetic risk and lifestyle with pancreatic cancer and their age dependency: a large prospective cohort study in the UK Biobank. *BMC Med*. 2023;21(1):489.
- Vujkovic M, Keaton JM, Lynch JA, Miller DR, Zhou J, Tcheandjie C, et al. Discovery of 318 new risk loci for type 2 diabetes and related vascular outcomes among 1.4 million participants in a multi-ancestry meta-analysis. *Nat Genet*. 2020;52(7):680–91.
- Zhuang P, Liu X, Li Y, Wan X, Wu Y, Wu F, et al. Effect of diet quality and genetic predisposition on hemoglobin A(1c) and type 2 diabetes risk: gene-diet interaction analysis of 357,419 individuals. *Diabetes Care*. 2021;44(11):2470–9.
- Sharma S, Tapper WJ, Collins A, Hamady ZZR. Predicting pancreatic cancer in the UK Biobank Cohort using polygenic risk scores and diabetes mellitus. *Gastroenterology*. 2022;162(6):1665–1674.e1662.
- Yang X, Kar S, Antoniou AC, Pharoah PDP. Polygenic scores in cancer. *Nat Rev Cancer*. 2023;23(9):619–30.
- Briggs SEW, Law P, East JE, Wordsworth S, Dunlop M, Houlston R, et al. Integrating genome-wide polygenic risk scores and non-genetic risk to predict colorectal cancer diagnosis using UK Biobank data: population based cohort study. *BMJ*. 2022;379: e071707.
- He Y, Lakhani CM, Rasooly D, Manrai AK, Tzoulaki I, Patel CJ. Comparisons of polyexposure, polygenic, and clinical risk scores in risk prediction of type 2 diabetes. *Diabetes Care*. 2021;44(4):935–43.
- Elliott J, Bodinier B, Bond TA, Chadeau-Hyam M, Evangelou E, Moons KGM, et al. Predictive accuracy of a polygenic risk score-enhanced prediction model vs a clinical risk score for coronary artery disease. *JAMA*. 2020;323(7):636–45.
- Clift AK, Tan PS, Patone M, Liao W, Coupland C, Bashford-Rogers R, et al. Predicting the risk of pancreatic cancer in adults with new-onset diabetes: development and internal-external validation of a clinical risk prediction model. *Br J Cancer*. 2024;130(12):1969–78.
- Cichosz SL, Jensen MH, Hejlesen O, Henriksen SD, Drewes AM, Olesen SS. Prediction of pancreatic cancer risk in patients with new-onset diabetes using a machine learning approach based on routine biochemical parameters. *Comput Methods Programs Biomed*. 2024;244: 107965.
- Boursi B, Finkelman B, Giantonio BJ, Haynes K, Rustgi AK, Rhim AD, et al. A clinical prediction model to assess risk for pancreatic cancer among patients with new-onset diabetes. *Gastroenterology*. 2017;152(4):840–850.e843.
- Khan S, Safarudin RF, Kupec JT. Validation of the ENDPAC model: identifying new-onset diabetics at risk of pancreatic cancer. *Pancreatol*. 2021;21(3):550–5.
- Sudlow C, Gallacher J, Allen N, Beral V, Burton P, Danesh J, et al. UK biobank: an open access resource for identifying the causes of a wide range of complex diseases of middle and old age. *PLoS Med*. 2015;12(3):e1001779.
- Bycroft C, Freeman C, Petkova D, Band G, Elliott LT, Sharp K, et al. The UK Biobank resource with deep phenotyping and genomic data. *Nature*. 2018;562(7726):203–9.
- Rashkin SR, Graff RE, Kachuri L, Thai KK, Alexeeff SE, Blatchins MA, et al. Pan-cancer study detects genetic risk variants and shared genetic basis in two large cohorts. *Nat Commun*. 2020;11(1):4423.
- Jia G, Lu Y, Wen W, Long J, Liu Y, Tao R, et al. Evaluating the utility of polygenic risk scores in identifying high-risk individuals for eight common cancers. *JNCI Cancer Spectr*. 2020;4(3):pkaa021.
- Nakatomi M, Lin Y, Ito H, Hara K, Kinoshita F, Kobayashi Y, et al. Prediction model for pancreatic cancer risk in the general Japanese population. *PLoS ONE*. 2018;13(9):e0203386.
- Galeotti AA, Gentiluomo M, Rizzato C, Obazee O, Neoptolemos JP, Pasquali C, et al. Polygenic and multifactorial scores for pancreatic ductal adenocarcinoma risk prediction. *J Med Genet*. 2021;58(6):369–77.

30. Sharma A, Kandlakunta H, Nagpal SJS, Feng Z, Hoos W, Petersen GM, et al. Model to determine risk of pancreatic cancer in patients with new-onset diabetes. *Gastroenterology*. 2018;155(3):730–739.e733.
31. White IR, Royston P. Imputing missing covariate values for the Cox model. *Stat Med*. 2009;28(15):1982–98.
32. Steyerberg EW, Harrell FE Jr. Prediction models need appropriate internal, internal-external, and external validation. *J Clin Epidemiol*. 2016;69:245–7.
33. Takada T, Nijman S, Denaxas S, Snell KIE, Uijl A, Nguyen TL, et al. Internal-external cross-validation helped to evaluate the generalizability of prediction models in large clustered datasets. *J Clin Epidemiol*. 2021;137:83–91.
34. Riley RD, Ensor J, Snell KI, Debray TP, Altman DG, Moons KG, et al. External validation of clinical prediction models using big datasets from e-health records or IPD meta-analysis: opportunities and challenges. *BMJ*. 2016;353:i3140.
35. Morris TP, White IR, Carpenter JR, Stanworth SJ, Royston P. Combining fractional polynomial model building with multiple imputation. *Stat Med*. 2015;34(25):3298–317.
36. Riley RD, Ensor J, Snell KIE, Harrell FE Jr, Martin GP, Reitsma JB, et al. Calculating the sample size required for developing a clinical prediction model. *BMJ*. 2020;368:m441.
37. Del Chiaro M, Sugawara T, Karam SD, Messersmith WA. Advances in the management of pancreatic cancer. *BMJ*. 2023;383: e073995.
38. Conroy T, Castan F, Lopez A, Turpin A, Ben Abdelghani M, Wei AC, et al. Five-year outcomes of FOLFIRINOX vs gemcitabine as adjuvant therapy for pancreatic cancer: a randomized clinical trial. *JAMA Oncol*. 2022;8(11):1571–8.
39. Hsieh MH, Sun LM, Lin CL, Hsieh MJ, Hsu CY, Kao CH. Development of a prediction model for pancreatic cancer in patients with type 2 diabetes using logistic regression and artificial neural network models. *Cancer Manag Res*. 2018;10:6317–24.
40. Chen W, Butler RK, Lustigova E, Chari ST, Wu BU. Validation of the enriching new-onset diabetes for pancreatic cancer model in a diverse and integrated healthcare setting. *Dig Dis Sci*. 2021;66(1):78–87.
41. van Alten S, Domingue BW, Faul J, Galama T, Marees AT. Reweighting UK Biobank corrects for pervasive selection bias due to volunteering. *Int J Epidemiol*. 2024;53(3):dyae054.

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